

B.Tech IV Year I Semester (R20) Regular Examinations December/January 2024

INTELLIGENT INFORMATION RETRIEVAL SYSTEMS

(Common to AI&DS, CSE (AI) and CSE (AI&ML))

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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| (a) Define information retrieval system. | 2M |
| (b) Describe the relationship between digital libraries and information retrieval systems. | 2M |
| (c) Describe the concept of information extraction with respect to indexing. | 2M |
| (d) Define indexing. Explain its significance in the information retrieval process. | 2M |
| (e) Explain the significance of the thesaurus. | 2M |
| (f) What is clustering? | 2M |
| (g) Explain the concept of interactive information retrieval. | 2M |
| (h) What is hypertext? Describe its role in searching and indexing. | 2M |
| (i) Differentiate between software text search algorithm and hardware text search algorithm. | 2M |
| (j) Explain in brief graph retrieval. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Examine various methods for evaluating the performance of information retrieval system. List and explain different metrics used for evaluation. 10M
- OR**
- 3 Explain the significance of feedback mechanism in improving information retrieval results. Also explain relevance feedback and user feedback. 10M
- 4 What is indexing? Explain in detail complete indexing process. Describe the impact of indexing on search results. 10M
- OR**
- 5 Define stemming. Explain the Porter stemmer algorithm. List its limitations. 10M
- 6 Explain the process of automatic term clustering. Illustrate its benefits in the information retrieval process. 10M
- OR**
- 7 What is statistical indexing? Describe its components. Illustrate its effect on the performance of a search process. 10M
- 8 Explain the concept of relevance feedback. Illustrate its impact in the performance of search process. 10M
- OR**
- 9 List the advantages and disadvantages of weighted search. Illustrate how weighted search enhances the effectiveness of the search algorithm. 10M
- 10 Discuss the importance of content-based retrieval in multimedia information retrieval. Explain the process of feature extraction and similarity measures in multimedia information systems. 10M
- OR**
- 11 Discuss in detail the evaluation measures used in the information retrieval system. Illustrate how these are calculated. 10M
